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An Explainable AI-based Deep Learning Model for Classification of Diabetic Retinopathy Stages Using Retinal Fundus Images

Samia Muntaha
Department of Computer Science and Engineering
Chittagong Independent University (CIU)
Chattogram, Bangladesh
muntaha.cse@ciu.edu.bd

Sayeda Sabiha Salam
Department of Electrical and Electronic Engineering
Chittagong Independent University (CIU)
Chattogram, Bangladesh
sabihasalam007@gmail.com

Naosher Mustakim
Department of Biomedical Physics and Technology
University of Dhaka
Dhaka, Bangladesh
naosher.mustakim@gmail.com

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***Abstract*— Diabetic Retinopathy (DR) is a vision-related illness in humans that causes retinal degeneration and can eventually lead to blindness. Diabetic retinopathy is one of the main causes of vision impairment globally and a major consequence of diabetes mellitus. Diabetic retinopathy detection in early stage is a must, in order to prevent total blindness. Numerous physical examinations, including visual clarity assessment, pupil dilation, and optical coherence tomography, are used to diagnose diabetic retinopathy; nevertheless, they are time-consuming and affect patients. This research outlines an innovative approach using Convolutional Neural Networks (CNN) for the detection of Diabetic Retinopathy (DR). This study builds a diversified dataset of retinal fundus images in order to detect five phases of Diabetic Retinopathy (DR) using a customized deep CNN architecture. After considerable research and parameter fine-tuning, the CNN outperformed existing approaches in terms of accuracy, sensitivity, and specificity in diagnosing DR stages. This study uses 3 publicly available datasets, previously imbalanced images across 5 classes, was oversampled to new images, resolving the imbalance issue. The suggested CNN architecture attained a testing accuracy of 95.25%, with testing precision, recall, and F1 score values of 96%, 95%, and 95%, respectively, across the three datasets. However, as a black-box model, its operation is difficult to explain, raising worries about the reliability of its results. To address this issue, we employed Gradient-Weighted Class Activation Map (Grad-CAM) for elucidation, which emphasizes significant areas within the photographs. This approach delivers classification results as well as insights, which help medical practitioners make decisions. An interpretability study revealed important visual regions for decision-making. This method not only advances diagnostic systems, but it also shows promise in early DR detection, potentially assisting in timely identification in clinical settings. It underscores the capabilities of convolutional neural networks (CNNs) in medical image processing, offering the prospect of enhanced accuracy in diabetic retinopathy (DR) screening, hence benefiting diabetic patients and diminishing healthcare expenses.**

Keywords*— *Diabetic Retinopathy, CNN, EyePACS, IDRiD

I. INTRODUCTION

Diabetic retinopathy (DR) is a progressive eye condition that impacts people with diabetes and is a leading cause of blindness among working-age adults. This results from persistent elevated blood glucose levels that impair delicate retinal blood vessels, leading to vision loss and even total blindness if not addressed. DR can go through two stages: non-proliferative diabetic retinopathy (NPDR) and proliferative diabetic retinopathy (PDR). There are four degrees of diabetic retinopathy severity: mild, moderate, severe, and proliferative [1]. Early detection of Diabetic Retinopathy (DR) is very important to prevent vision loss and promote overall well-being of patients suffering from diabetes. Deep learning methods, especially with CNNs has led to a paradigm shift towards achieving an efficient and also scalable strategy for DR detection which could potentially transform the early screening process and aids in managing Diabetic Retinopathy. Conversely, manual diabetic retinopathy screening is costly and labor-intensive, making early detection of the condition challenging. To solve this issue, several researchers have proposed different methods for DR detection, one of which utilizes neural networks to classify stages of diabetic retinopathy [2] [3]. Four techniques utilizing pre-trained models were proposed: VGG16, VGG19, InceptionV3, and DenseNet169. [4]. One of these studies was proposed in]. Some researchers have proposed deep learning techniques, particularly CNN [5]. A study presented a transfer learning-based Convolutional Neural Network (CNN) method for multi-class, multi-label classification of ophthalmic disorders utilizing fundus pictures. [6]. Another study suggested a hybrid method for the identification and

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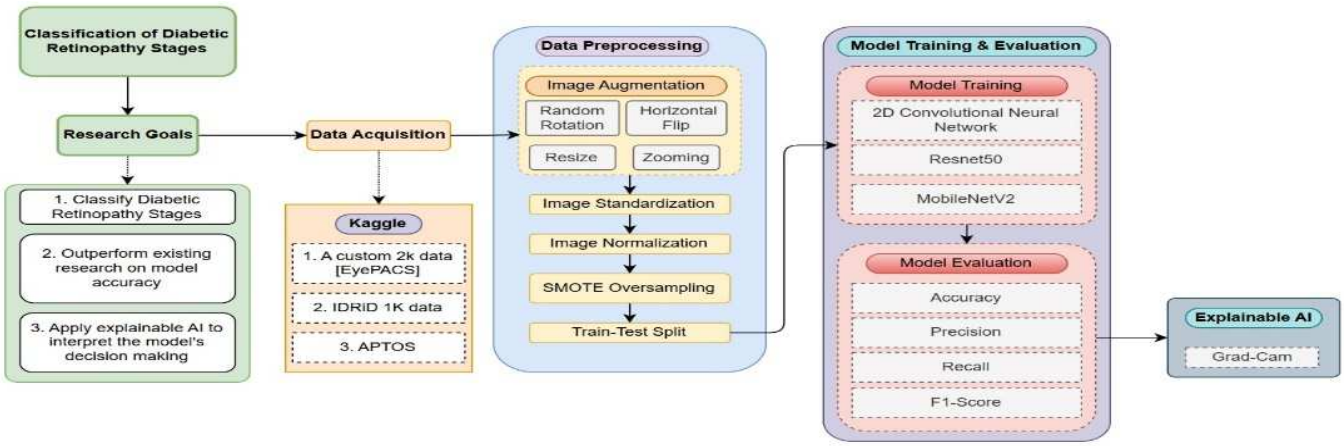

Fig. 1. Methodology diagram of the proposed approach.

classification of diabetic retinopathy using fundus eye pictures that combines transfer learning and pre-trained CNN models. The approach utilizes a combined feature vector, which is then fed into multiple classifiers for both binary and multiclass categorization. [7]. While CNN, have been widely employed for DR detection, most studies rely on huge, unbalanced datasets, frequently larger than 10,000 images. As a result, models may become biased, giving preference to majority classes and underperforming on minority ones. A study that classified diabetic retinopathy using the EyePACS dataset, which includes 71,049 pictures, produced multiclass classification accuracy of less than 80% [8]. One of the studies enhances CNN for Diabetic Retinopathy detection using Grad-CAM[9].

To our knowledge, no previous research has used 2D CNN architectures and short, balanced datasets to study DR detection. In this work, we try to present a novel approach that uses a balanced dataset of 2,000 photos over five DR classes to reach a high accuracy of 95.25%. This outcome highlights the importance of balanced dataset by outperforming earlier models trained on much bigger datasets. Our results demonstrate that quality datasets can improve classification accuracy more than quantity.

Deep learning is a precise, non-invasive and affordable method. CNNs have exhibited significant accomplishments in diverse image recognition assignments, suggesting that the creation of a CNN-centered system could aid in diagnosis, offering swift and precise outcomes. Our objective is to deploy an efficient model with high accuracy for the detection of diabetic retinopathy (DR). Utilizing a deep learning model may enhance individuals' quality of life by facilitating prompt identification and rapid treatment. This study aims to develop a CNN-based system for early, accurate Diabetic Retinopathy detection, develop an accessible diagnostic website, and contribute to medical image analysis using AI, ultimately enhancing DR diagnosis and healthcare accessibility. To ensure suitability, the dataset was preprocessed before model training. After examining different parameter values, a deep learning model for the identification of diabetic retinopathy was shown to be effective.

II. METHODS

This section describes the overall process of our work including dataset preparation, data preprocessing, model training and implementation of explainable AI techniques on the model. The methodology of our work can be explained using fig. 1.

A. Datasets

This study utilized three publicly available datasets. The first one, a custom 2k dataset, was sourced from the Kaggle EyePACS dataset [10]. The second dataset, IDRiD 1K, was collectd from the Indian Diabetic Retinopathy Image Dataset and consists of 516 retinal images. [11]. The third dataset, containing 6,282 retinal images, was collected from the APTOS Dataset [12]. Each dataset is organized into five distinct directories that categorize photos depending on the severity of diabetic retinopathy. They are as follows:

0 - No_DR
1 - Mild
2 - Moderate
3 - Severe
4 - Proliferate_DR

B. Dataset Preprocessing

Preprocessing this data was essential for increasing the ability of our deep learning model to identify diabetic retinopathy. Important steps included data augmentation, image standardization, and normalization. Image resizing ensured uniform input dimensions, while normalization scaled pixel values to [0, 1]. The resilience of the model was further improved using data augmentation methods such random rotations, flips, and zooming.

In this case there was a significant class imbalance present among the different stages of diabetic retinopathy. The amount of data in No DR was almost 5 times bigger than

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the data in Proliferative DR in Kaggle EyePACS and APTOS dataset and also on IDRiD dataset. Due to class imbalance, oversampling was employed to address the disparity, ensuring adequate representation of all severity levels during model training. SMOTE oversampling technique is applied in this case. We balanced the APTOS dataset by oversampling and created dataset with a total of 6879 images, but each class for training and testing did not contain the same number of images due to the classes being excessively imbalanced. From the EyePACS dataset very consciously we collected 2000 images for a balanced DR dataset. After applying oversampling on IDRiD dataset we made the balanced 1000 image dataset. For performance evaluation of the model on untested data, the datasets were divided into 80% training and 20% testing subsets. This comprehensive preprocessing allowed for effective training and evaluation, crucial for accurate diabetic retinopathy detection.

C. *Proposed 2D CNN Model*

Initially developed for object identification [15, 16] and picture classification [13, 14], CNN is now extensively used in biomedical signal analysis.

2D Convolutional Neural Network (CNN) was created for custom image datasets, using Keras for data preparation, image loading, resizing, display, data augmentation using ImageDataGenerator, and creating data generators for training and testing. The generators are configured with a target size of (128, 128) for image dimensions, a batch size of 32 images, and "rgb" color mode. These steps prepare augmented image batches and labels for model training and testing, enhancing the model's generalization. Fig. 2 depicts the general architecture of the suggested CNN model.

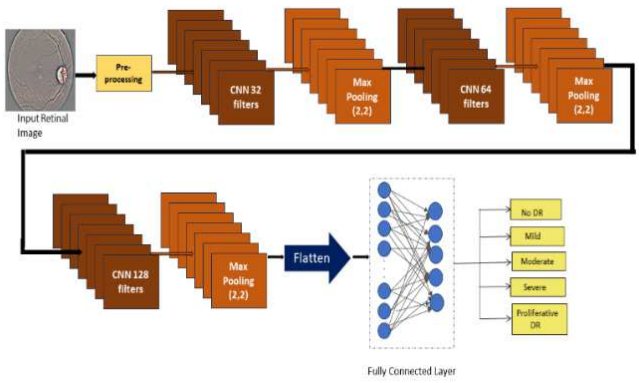

Fig. 2. Architecture of the 2D CNN model.

Three Convolutional Layers with different filter sizes (3,3) make up the model architecture, and each one applies a ReLU activation function. Three MaxPooling Layers use the MaxPooling2D function with a preset pool size of (2,2). A flattening layer is used to convert the output into a 1D array. Two dense Layers with 512 neurones and ReLU activation, five neurones for classification, and a 0.5-rate dropout layer are then added. Activation function or fully connected layers make up the classifier, which produces probabilities for dividing images into their classes. With a learning rate of 0.0001, the model was assembled using the Adam optimiser, and the loss function was categorical cross-entropy.

D. *ResNet (Residual Neural Networks)*

A major advancement in deep learning and convolutional neural networks (CNNs) is ResNet, or Residual Neural Networks. It was created by Kaiming He et al. to help with problems like vanishing gradients that arise when training extremely deep networks and to make training models with hundreds or thousands of layers possible. Performance of traditional deep networks was hindered as they got deeper due to vanishing gradients. ResNet's novel approach introduces residual blocks, in which networks learn to distinguish between desired outputs and inputs, allowing for easier optimization and the construction of much deeper networks. The architecture of ResNet is built by stacking several residual blocks, each of which has convolutional layers, batch normalisation, ReLU activation functions, and skip connections to facilitate gradient flow. For image classification applications, variants like ResNet-50, ResNet-101, and ResNet-152—which include between 50 and more layers—are frequently utilised. For training purposes, we used ResNet-50 in our model.

E. *MobileNetV2 (Mobile Network version 2)*

While MobileNetV2 and the original MobileNet are very similar, it employs bottleneck characteristics and inverted residual blocks to increase efficiency. The output of this function is a Keras image classification model that can have ImageNet pretrained weights inserted into it [17]. Effective transfer learning for multiple image classification tasks is facilitated by pre-training on ImageNet and then fine-tuning. This study used MobileNetV2 which we pre-trained on ImageNet. It does not have a top classification layer. Added a top layer with a dropout layer for regularisation, a dense layer with 512 ReLU units, a global average pooling layer, and another dense layer with a softmax activation function to classify multiple classes. During the first training, the base MobileNetV2 layers are frozen to increase training efficiency. As the metric, we employ the Adam optimiser with categorical cross-entropy loss and accuracy.

F. *Grad-CAM (Gradient-Weighted Class Activation Mapping)*

Grad-CAM clarifies the CNN model by pinpointing the regions of the image that the model is concentrating on. Again, it allows to test and validate the model by indicating

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which parts of the image contain the most significant information for classification, as well as gain a better understanding of how the model makes decisions [18]. This study utilized Convolutional Neural Networks (CNN) along with Gradient-Weighted Class Activation Mapping (Grad-CAM) to visually interpret the features in diabetic retinopathy images.

III. RESULTS

a) Results for CNN

Dataset 1- IDRiD Dataset with 1000 balanced Images: The model trained on the initial dataset (IDRiD) including 1,000 images attained an accuracy of 75%. The confusion matrices and classification reports demonstrate precision, recall, and F1 scores for five classes.

Dataset 2 - Custom Dataset with 2,000 Balanced Images from EyePACS: On the custom-balanced dataset of 2,000 images, the model achieved a significantly higher accuracy of 95.25%. The confusion matrix showed more balanced classification across all DR stages, indicating that the balanced dataset allowed the model to generalize better and reduce bias towards majority classes.

Dataset 3 - APTOS Dataset: On the APTOS dataset of 6,879 images, the model achieved accuracy of 88.00%. The confusion matrix for this dataset showed that the model struggled to classify the minority DR classes, which was most likely due to the dataset's class imbalance.

Fig. 3, 4 and 5 show the confusion matrices of the suggested CNN model on these three datasets. Fig. 6 illustrates the training and testing accuracy curves for 2K Dataset. To assess the proposed methods, various indices were calculated, including accuracy, sensitivity, specificity, and F1-score. The metrics were calculated using formulas (1-4), where TN denotes true negatives, TP suggests true positives, FN represents false negatives, and FP shows false positives. The equations are as follows:

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (1)$$

$$Precision = \frac{TP}{(TP + FP)} \quad (2)$$

$$Sensitivity\ or\ Recall = \frac{TP}{(TP + FN)} \quad (3)$$

$$F1 - score = \frac{2 \times (Precision \times Recall)}{(\text{Precision} + \text{Recall})} \quad (4)$$

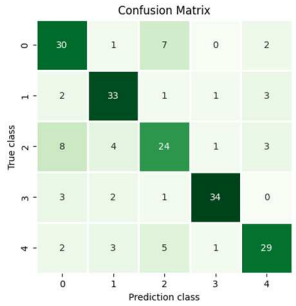

Fig. 3. Confusion matrix of the 2D CNN model (IDRiD Dataset).

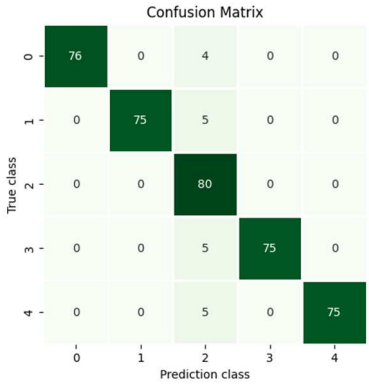

Fig. 4 Confusion matrix of the 2D CNN model (Custom Balanced Dataset).

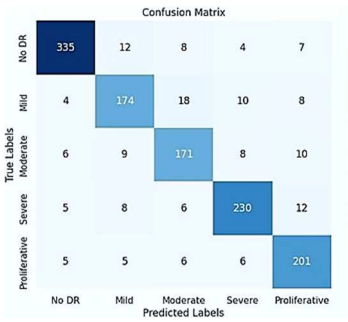

Fig. 5. Confusion matrix of the 2D CNN model (APTOS Dataset).

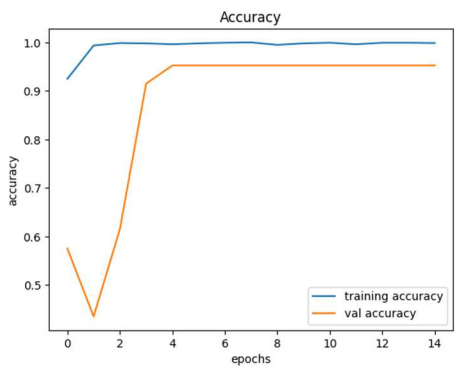

Fig. 6. Training and Validation Accuracy Curve for CNN Model

b) Other Model's Performance

The accuracy, precision, recall and f1-score for ResNet-50 are 85%, 86%, 85% and 86% respectively and for

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MobileNetV2 all are 84% on our custom dataset. Fig. 7 and fig.8 show the confusion matrices of ResNet-50 and MobileNetV2 respectively.

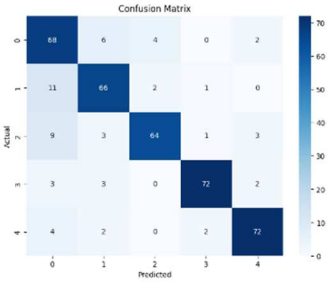

Fig. 7. Confusion Matrix for ResNet-50.

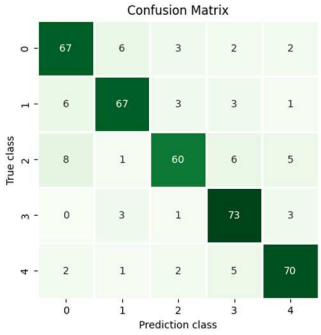

Fig. 8. Confusion Matrix for MobileNetV2.

Fig. 9(a) shows the Grad-Cam results along with the original fundus image. Green to Yellow Areas indicate the most influential areas for the model's decision-making process. Here the model focuses more on abnormal retinal regions, like micro aneurysms, hemorrhages, or other pathological features associated with DR as expected. The model focuses on a broad area in the upper-left and central parts of the retina in the first image of fig. 8, including the optic disc. While the optic disc is a natural feature, the involvement of surrounding areas could indicate that the model is looking at potential lesions.

Fig 9(b) shows more concentrated attention on a small region toward the top right of the image, potentially suggesting a localized anomaly like exudates or small hemorrhages. The model has narrowed down on a specific area, which is promising if that area shows diabetic retinopathy signs. In Fig. 9(c), the attention is spread across multiple areas, with a significant focus in the top left and central parts of the image. This could suggest the presence of several areas of interest that the model is interpreting as relevant to the condition.

a) Performance Comparison (CNN)

Machine learning models aim to detect patterns in data by learning from training sets and then applying their findings to previously unseen test data. It is critical to ensure the model's efficacy on new data. Accuracy measures overall accuracy, precision examines category-specific prediction quality, recall assesses a model's capacity to identify specific categories, and F1-score serves as a vital evaluation metric in machine learning. Table I presents the performance metrics for the CNN model across various datasets.

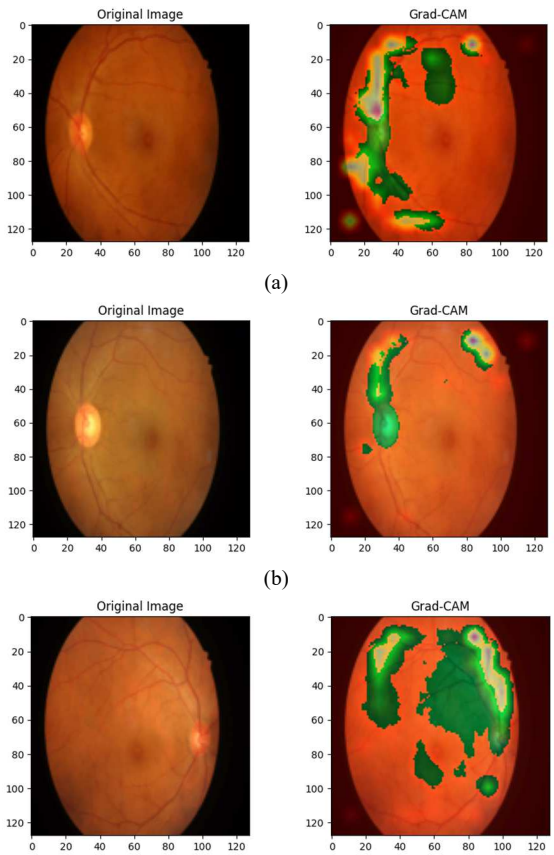

Fig. 9(a),9(b),9(c). Grad-Cam results along with the original fundus images.

TABLE I. PERFORMANCE METRICS FOR DIFFERENT DATASETS

2D CNN Model Performance						
Dataset	***Epochs***	***Accuracy Training (%)***	***Accuracy Validation(%)***	***Precision (%)***	***Recall (%)***	***F1-score (%)***
IDRiD(1K)	40	80.47	75.00	75.00	75.00	75.00
APTOS (Unbalanced)	30	95.20	88.00	87.00	87.00	87.00
Custom 2K	15	99.87	95.25	96.00	95.00	95.00

Fig. 10 presents the accuracies for the CNN model across various datasets. It is evident from our experiments that as the size of the balanced dataset size increases from 1000 to 2000, the accuracy improves from 75.00% to 95.25%. Again, with APTOS imbalanced dataset the accuracy shows 88%. This finding demonstrates a positive relationship between size of a balanced dataset and accuracy, implying that balanced datasets contribute to improved prediction accuracy of the model. Table II

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presents the performance metrics for the Custom 2K dataset across various models (MobileNetV2, ResNet-50, and 2D CNN).

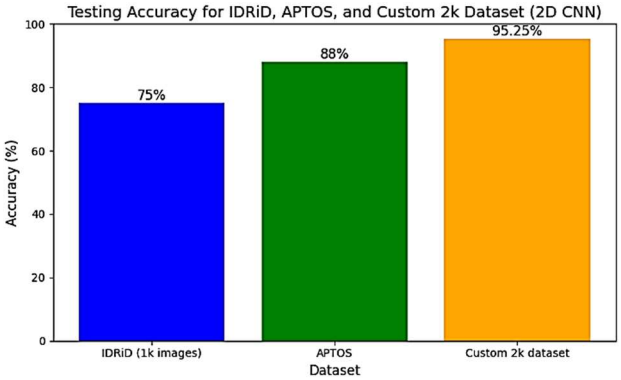

Fig. 10. Testing Accuracies of three datasets.

Tale II. Performance Metrics for Different Models on Custom 2K dataset

Model Performance						
Deep L,/ML Model	***Epochs***	***Accuracy Training(%)***	***Accuracy Validation(%)***	***Precision (%)***	***Recall (%)***	***F1-score (%)***
MobileNetV2	35	97.58	84.25	84.00	84.00	84.00
ResNet-50	25	98.69	85.50	86.00	85.00	86.00
2D CNN	15	99.87	95.25	96.00	95.00	95.00

IV. CONCLUSIONS

The suggested method aims to establish a system for detecting diabetic retinopathy in diabetic patients, facilitating ophthalmologists in the early identification of its symptoms. Compared to previous research, this study shows a significant improvement in accuracy, particularly on the small custom-balanced sample. The addition of a more balanced training set, as well as data augmentation approaches, significantly improved the model's robustness and performance. The primary contributions of this work include: A small custom balanced dataset was collected to produce a valid and well-balanced diabetic retinopathy (DR) detection dataset with five classes. Pre-trained convolutional neural networks (CNNs), specifically ResNet50 and MobileNet, were selected based on their performance evaluations in prior studies. A Grad-CAM visual explanation method was used to debug the prediction process of the 2D CNN model and highlight specific points in the diabetic retinopathy images that influenced the final decision. The experimental findings show that the 2D CNN model surpasses both ResNet50 and MobileNetV2. On a balanced sample, our model outperformed standard approaches, achieving 95.25% accuracy.

V. REFERENCES

[1] Hamdy, F. C., Donovan, J., Lane, J. A., Mason, M. D., Metcalfe, C., Holding, P., Davis, M. J., Peters, T. J., Turner, E. L., Martin, R. M., Oxley, J., Robinson, M. C., Staffurth, J., Walsh, E., Bollina, P., Catto, J. W., Doble, A., Doherty, A., Gillatt, D., . . . Neal, D. E. (2016c). 10-Year Outcomes after Monitoring, Surgery, or Radiotherapy for Localized Prostate Cancer. The New England Journal of Medicine, 375(15), 1415–1424. https://doi.org/10.1056/nejmoa1606220

[2] Abràmoff M.D., Lou Y., Erginay A., et al. (2016). "Detection of Diabetic Retinopathy using Convolutional Neural Networks." Proceedings of the National Academy of Sciences of the United States of America. DOI: 10.1073/pnas.1609122113.

[3] Rajalakshmi R., Subramaniam G., et al. (2018). "Deep Learning for Automated Diabetic Retinopathy Detection using Retinal Fundus Images." Medical Image Analysis. DOI:10.1016/j.media.2018.09.008.

[4] Kallel, F., & Echtioui, A. (2023). Retinal fundus image classification for diabetic retinopathy using transfer learning technique. Signal, Image and Video Processing, 1-11.

[5] Latif, G., Bouchard, K., Maitre, J., Back, A., & Bédard, L. P. (2022). Deep-learning-based automatic mineral grain segmentation and recognition. Minerals, 12(4), 455.

[6] Gour, N., & Khanna, P. (2021). Multi-class multi-label ophthalmological disease detection using transfer learning based convolutional neural network. *Biomedical Signal Processing and Control*, *66*, 102329. https://doi.org/10.1016/j.bspc.2020.102329

[7] Butt, M. M., Iskandar, D., Abdelhamid, S., Latif, G., & Alghazo, R. (2022). Diabetic Retinopathy Detection from Fundus Images of the Eye Using Hybrid Deep Learning Features. *Diagnostics*, *12*(7), 1607. https://doi.org/10.3390/diagnostics12071607

[8] Biswas, A., & Banik, R. (2023). Advance Detection of Diabetic Retinopathy: Deep Learning approach. In *Communications in computer and information science* (pp. 64–77). https://doi.org/10.1007/978-3-031-48876-4_6

[9] Duvvuri, K., Chethana, S., Charan, S. S., Srihitha, V., Ramesh, T. K., & S, S. K. (2022). Grad-CAM for Visualizing Diabetic Retinopathy. 2022 3rd International Conference for Emerging Technology (INCET). https://doi.org/10.1109/incet54531.2022.9824598

[10] *eyepacs*. (2024, April 29). Kaggle. https://www.kaggle.com/datasets/andreivann/eyepacs

[11] Prasanna Porwal, Samiksha Pachade, Ravi Kamble, Manesh Kokare, Girish Deshmukh, Vivek Sahasrabuddhe, Fabrice Meriaudeau, April 24, 2018, "Indian Diabetic Retinopathy Image Dataset (IDRiD)", IEEE Dataport, doi: https://dx.doi.org/10.21227/H25W98.

[12] Karthik, Maggie, & Sohier Dane. (2019). APTOS 2019 Blindness Detection. Kaggle. https://kaggle.com/competitions/aptos2019-blindness-detection

[13] Bharadiya, J. "Convolutional neural networks for image classification." International Journal of Innovative Science and Research Technology 8.5 (2023): 673-677.

[14] Y. LeCun et al., "Backpropagation applied to handwritten zip code recognition," Neural computation, vol. 1, no. 4, pp. 541-551, 1989

[15] S. R. Sitaraman, M. V. S. Narayana, J. Lande, L. M and A. H. Shnain, "Center Intersection of Union loss with You Only Look Once for Object Detection and Recognition," 2024 International Conference on Intelligent Algorithms for Computational Intelligence Systems (IACIS), Hassan, India, 2024, pp. 1-4, doi: 10.1109/IACIS61494.2024.10721907.

[16] Xu, J., Ren, H., Cai, S., & Zhang, X. (2022). An improved faster R-CNN algorithm for assisted detection of lung nodules. Computers in Biology and Medicine, 153, 106470. https://doi.org/10.1016/j.compbiomed.2022.106470

[17] Team, K. (n.d.). *Keras documentation: MobileNet, MobileNetV2, and MobileNetV3*. https://keras.io/api/applications/mobilenet/

[18] Moujahid, H., Cherradi, B., Al-Sarem, M., Bahatti, L., Eljialy, A. B. a. M. Y., Alsaeedi, A., & Saeed, F. (2021). Combining CNN and Grad-Cam for COVID-19 disease prediction and visual explanation. *Intelligent Automation & Soft Computing*, *32*(2), 723–745. https://doi.org/10.32604/iasc.2022.022179

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